

COMP-SCI-5565

Intro to Statistical Learning

Topic 5 – Linear Model Selection and Regularization

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Lecture Objectives

- Compare and contrast subset selection methods – forward stepwise, backward stepwise, hybrid, and best subset.
- Use shrinkage methods to constrain the flexibility of linear models.
- Compare and contrast the lasso and ridge regression methods of shrinkage.
- Reduce the dimensionality of the data for a linear model.
- Compare and contrast the PCR and PLS methods of dimension reduction.
- Explain the challenges that may occur when fitting linear models to high-dimensional data.

Outline

5.1 Introduction

5.2 Subset Selection

5.3 Shrinkage Methods

5.4 Dimension Reduction Methods

5.5 Considerations in High Dimensions

Outline

5.1 Introduction

5.2 Subset Selection

5.3 Shrinkage Methods

5.4 Dimension Reduction Methods

5.5 Considerations in High Dimensions

5.1 Introduction

- Recall the linear model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- Despite its simplicity, the linear model has distinct advantages in terms of its **interpretability** and often shows good **predictive performance**.
- Hence, we discuss in this lecture some ways in which the multiple linear model can be improved, by removing or constraining predictors.

5.1 Introduction

Why constrain or remove predictors?

- **Prediction Accuracy**: especially when $p > n$, to control the variance. Or meaningless $p = n$.
Such situations cause high variance.
- **Model Interpretability**: remove or constrain irrelevant variables to simplify model.
By removing irrelevant features – that is, by setting the corresponding coefficient estimates to zero.
We can obtain a model that is more easily interpreted.

5.1 Introduction

Three classes of methods

- **Subset Selection.** We identify a subset of the p predictors related to the response. We then fit a model using least squares on the reduced set of variables.
- **Shrinkage.** We fit a model involving all p predictors, but the estimated coefficients are shrunk towards zero relative to the least squares estimates. This shrinkage (also known as regularization) reduces variance and can also perform variable selection.
- **Dimension Reduction.** We project the p predictors into a M -dimensional subspace, where $M < p$.

Outline

5.1 Introduction

5.2 Subset Selection

5.3 Shrinkage Methods

5.4 Dimension Reduction Methods

5.5 Considerations in High Dimensions

5.2 Subset Selection

- This group of methods directly reduces the number of predictors by removing the ones that don't improve the test error.

Best Subset Selection (BSS)

- Most straightforward approach - try them all!
- We fit a separate least squares regression for each possible combination of the p predictors.
- That is, we fit all p models selection that contain exactly one predictor, all $\binom{p}{2} = p(p-1)/2$ models that contain exactly two predictors, and so forth.

5.2 Subset Selection

Best Subset Selection (BSS) – continued

BSS Algorithm

1. Start with the null model (intercept-only model), $\mathcal{M}_0$
2. For $k = 1, 2, \dots, p$:
 - 2.1 Fit all $\binom{p}{k}$ models containing k predictors.
 - 2.2 Let $\mathcal{M}_k$ denote the best of these $\binom{p}{k}$ models, where best is defined as having the lowest RSS, lowest deviance, etc.
3. Choose the best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$ where best is defined as having the lowest C_p , BIC , AIC , cross-validation MSE, or, alternatively, highest adjusted R^2 .

5.2 Subset Selection

Best Subset Selection (BSS) – continued



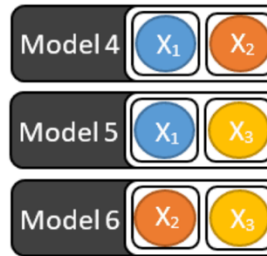
Step 1: Consider All Possible Models

By listing all possible combination of variables

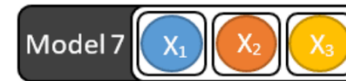
Models with 1 variable:



Models with 2 variables:



Models with 3 variables:



Step 2: Identify the Best Model of Each Size

By choosing the one with the lowest sum of squared errors or the highest R^2

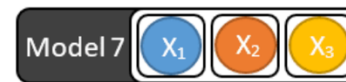
Best model with 1 variable:



Best model with 2 variables:



Best model with 3 variables:



Step 3: Identify the Best Overall Model

By choosing the one with the lowest AIC (or BIC) or the highest adjusted R^2

Best overall model:



5.2 Subset Selection

Best Subset Selection (BSS) – continued

Pros

- Selects the best subset

Cons

- Overfitting due to large search space.
- Computationally expensive, intractable for large p (exponential, 2^p , e.g., $p = 20$ yields over 1 million possibilities).

5.2 Subset Selection

Forward Stepwise Subset Selection (FsSS)

- Forward stepwise selection begins with a model containing no predictors and then adds predictors to the model, one at a time, until all of the predictors are in the model.
- In particular, at each step, the variable that gives the greatest additional improvement to the fit is added to the model.

5.2 Subset Selection

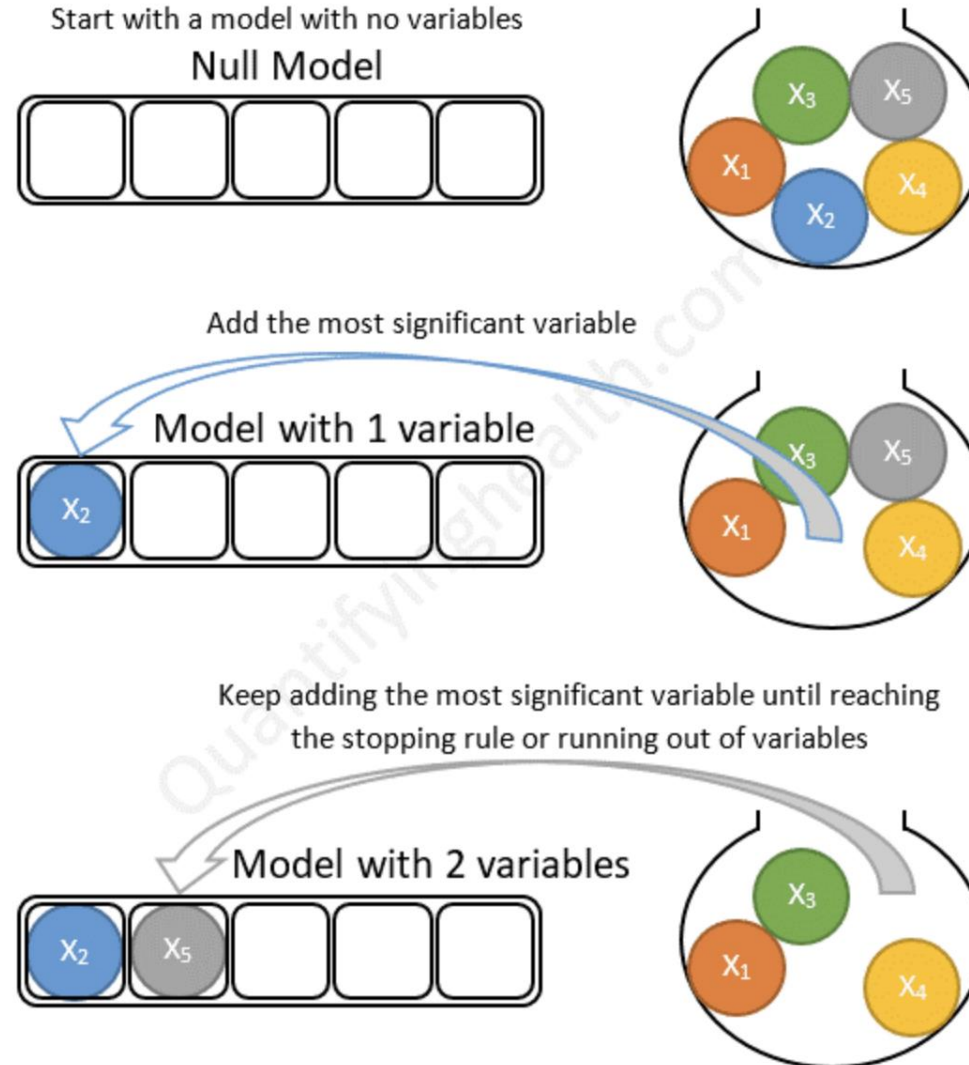
Forward Stepwise Subset Selection (FsSS) – continued

FsSS Algorithm

1. Let $\mathcal{M}_0$ denote the null model, which contains no predictors.
2. For $k = 0, \dots, p - 1$:
 - 2.1 Consider all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
 - 2.2. Choose the best among these $p - k$ models, and call it $\mathcal{M}_{k+1}$. Here, best is defined as having the smallest RSS or highest R^2 .
3. Choose the best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$ where best is defined as having the lowest C_p , BIC , AIC , cross-validation MSE, or, alternatively, highest adjusted R^2 .

5.2 Subset Selection

Forward Stepwise Subset Selection (FsSS) – continued



5.2 Subset Selection

Forward Stepwise Subset Selection (FsSS) – continued

Pros

- Computational advantage over best subset selection is clear.

Cons

- Finding the best possible model out of all 2^p models containing subsets of the p predictors is not guaranteed. Why not?

5.2 Subset Selection

Backward Stepwise Subset Selection (BsSS)

- Like forward stepwise selection, backward stepwise selection provides an efficient alternative to best subset selection.
- However, unlike forward stepwise selection, it begins with the full least squares model containing all p predictors and then iteratively removes the least useful predictor, one-at-a-time.

5.2 Subset Selection

Backward Stepwise Subset Selection (BsSS) - continued

BsSS Algorithm

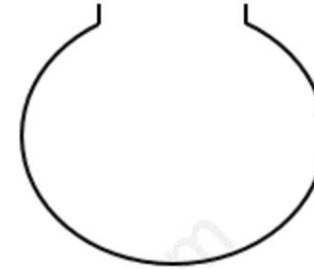
1. Let $\mathcal{M}_0$ denote the full model, which contains all predictors.
2. For $k = p, p - 1, \dots, 1$:
 - 2.1 Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k - 1$ predictors.
 - 2.2. Choose the best among these k models and call it $\mathcal{M}_{k-1}$. Here, best is defined as having the smallest RSS or highest R^2 .
3. Choose the best model among $\mathcal{M}_0, \dots, \mathcal{M}_p$ where best is defined as having the lowest C_p , BIC , AIC , cross-validation MSE, or, alternatively, highest adjusted R^2 .

5.2 Subset Selection

Backward Stepwise Subset Selection (BsSS) - continued

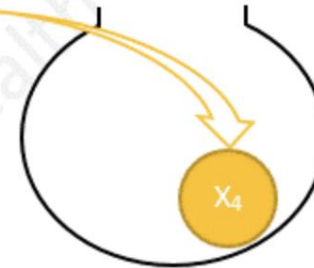
Start with a model that contains all the variables

Full Model



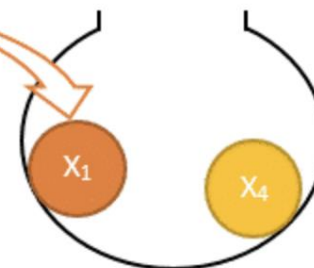
Remove the least significant variable

Model with 4 variables



Keep removing the least significant variable until reaching the stopping rule or running out of variables

Model with 3 variables



5.2 Subset Selection

Backward Stepwise Subset Selection (BsSS) - continued

Pros

- Computational advantage over best subset selection is clear.
- Starting with the full model has the advantage of considering the effects of all variables simultaneously.

Cons

- Like forward stepwise, finding the best possible model out of all 2^p models containing subsets of the p predictors is not guaranteed. Why not?
- Backward selection requires that the number of samples n is larger than the number of variables p (so that the full model can be fit).
- In contrast, forward stepwise can be used even when $n < p$, and so is the only viable subset method when p is very large.

5.2 Subset Selection

Hybrid searches

- Combine FsSS and BsSS
- Variables are added sequentially, but remove any variables that no longer improve fit after adding.

Outline

5.1 Introduction

5.2 Subset Selection

5.3 Shrinkage Methods

5.4 Dimension Reduction Methods

5.5 Considerations in High Dimensions

5.3 Shrinkage Methods

Overview

- The subset selection methods use least squares to fit a linear model that contains a subset of the predictors.
- Alternatively, we can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates or shrinks the coefficient estimates toward zero.
- It may not be immediately obvious why such a constraint should improve the fit, but shrinking the coefficient estimates can significantly reduce their variance.

5.3 Shrinkage Methods

Ridge Regression

- Ridge regression—also known as L2 regularization—is one of several types of regularization for linear regression models.
- Ridge regression specifically corrects for **multicollinearity** in regression analysis.
- **Multicollinearity** is when two or more predictors have a near-linear relationship.
- This is also useful when dealing with many parameters, particularly if those parameters also have high weights.
- While this lecture focuses on regularizing linear regression models, ridge regression may also be applied in logistic regression.

5.3 Shrinkage Methods

Ridge Regression – continued

- Recall that the least squares fitting procedure estimates $\beta_0, \beta_1, \dots, \beta_p$ using the values that minimize

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- If $\text{RSS} = 0$, the model perfectly predicts dependent variables.
- A score of zero is not always desirable – it can indicate overfitting on the training data.
- High coefficient estimates can often indicate overfitting, and the model's output becomes sensitive to minor alterations in the input data.

5.3 Shrinkage Methods

Ridge Regression – continued

- Ridge regression corrects for high-value coefficients by introducing a regularization term (often called the penalty term) into the RSS function.
- This penalty term is the sum of the squares of the model's coefficients.

$$L2 = \|\beta\|_2 = \beta_1^2 + \beta_2^2 + \dots + \beta_p^2$$

- The L2 penalty term is inserted as the end of the RSS function, resulting in a new formulation of RSS.

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

where $\lambda \geq 0$ is a tuning parameter to control its effect on the model, to be determined separately using cross-validation.

5.3 Shrinkage Methods

Ridge Regression – continued

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

- Remember that coefficients mark a given predictor's (i.e. independent variable's) effect on the predicted value (i.e., dependent variable).
- Once added into RSS formula, the L2 penalty term counteracts especially high coefficients by reducing all coefficient values.
- The ridge estimator thus calculates new regression coefficients that reduce a given model's RSS.
- This minimizes every predictor's effect and reduces overfitting on training data.

5.3 Shrinkage Methods

Ridge Regression – continued

- Note that ridge regression does not shrink every coefficient by the same value.
- Rather, coefficients are shrunk in proportion to their initial size.
- As λ increases, high-value coefficients shrink more than low-value coefficients.
- High-value coefficients are thus penalized greater than low-value coefficients.
- Ridge regression does have one obvious disadvantage: unlike subset selection, which will generally select models that involve just a subset of the variables, ridge regression will include all p predictors in the final model

5.3 Shrinkage Methods

The Lasso

- Ridge regression has one obvious disadvantage: unlike subset selection, which generally selects models that involve just a subset of the variables, ridge regression will include all p predictors in the final model.
- The Lasso is a relatively recent alternative to ridge regression that overcomes this disadvantage.
- The acronym “LASSO” stands for Least Absolute Shrinkage and Selection Operator.
- Lasso regression performs L1 regularization, which adds a penalty equal to the absolute value of the magnitude of coefficients.

5.3 Shrinkage Methods

The Lasso – continued

- L1 regularization

$$L1 = \|\beta\|_1 = \beta_0 + \beta_1 + \cdots + \beta_p$$

- RSS becomes

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

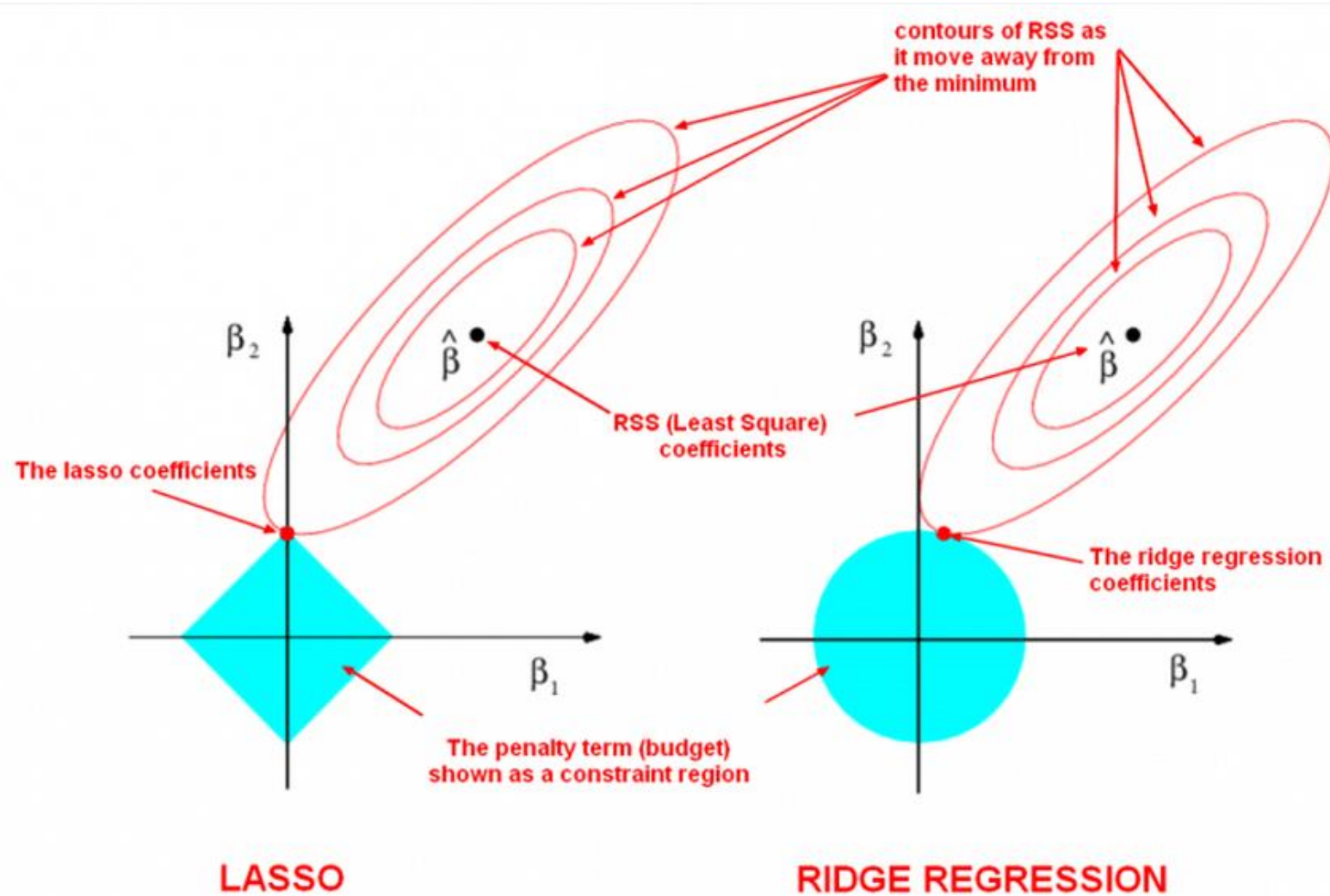
where $\lambda \geq 0$ is a tuning parameter to control its effect on the model, to be determined separately using cross-validation.

5.3 Shrinkage Methods

The Lasso – continued

- Lasso can result in sparse models with few coefficients.
- Some coefficients can become zero and eliminated from the model.
- Larger penalties, λ , result in coefficient values closer to zero, which is ideal for producing simpler models.
- On the other hand, Ridge regression doesn't result in the elimination of coefficients or sparse models.
- This makes the Lasso far easier to interpret than the Ridge.

5.3 Shrinkage Methods



Outline

5.1 Introduction

5.2 Subset Selection

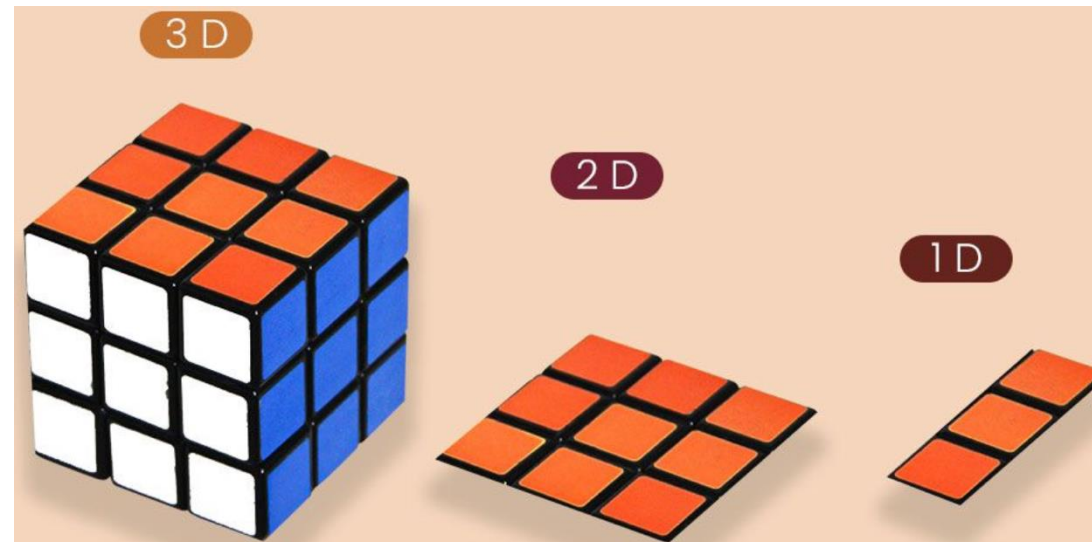
5.3 Shrinkage Methods

5.4 Dimension Reduction Methods

5.5 Considerations in High Dimensions

5.4 Dimension Reduction Methods

- The methods we have discussed so far involved fitting linear regression models via least squares or a shrunk approach using the original predictors, $X_1, X_2, \dots, X_p$.
- We now explore a class of approaches that transform the predictors and then fit a least squares model using the transformed variables.
- We will refer to these techniques as dimension reduction methods.



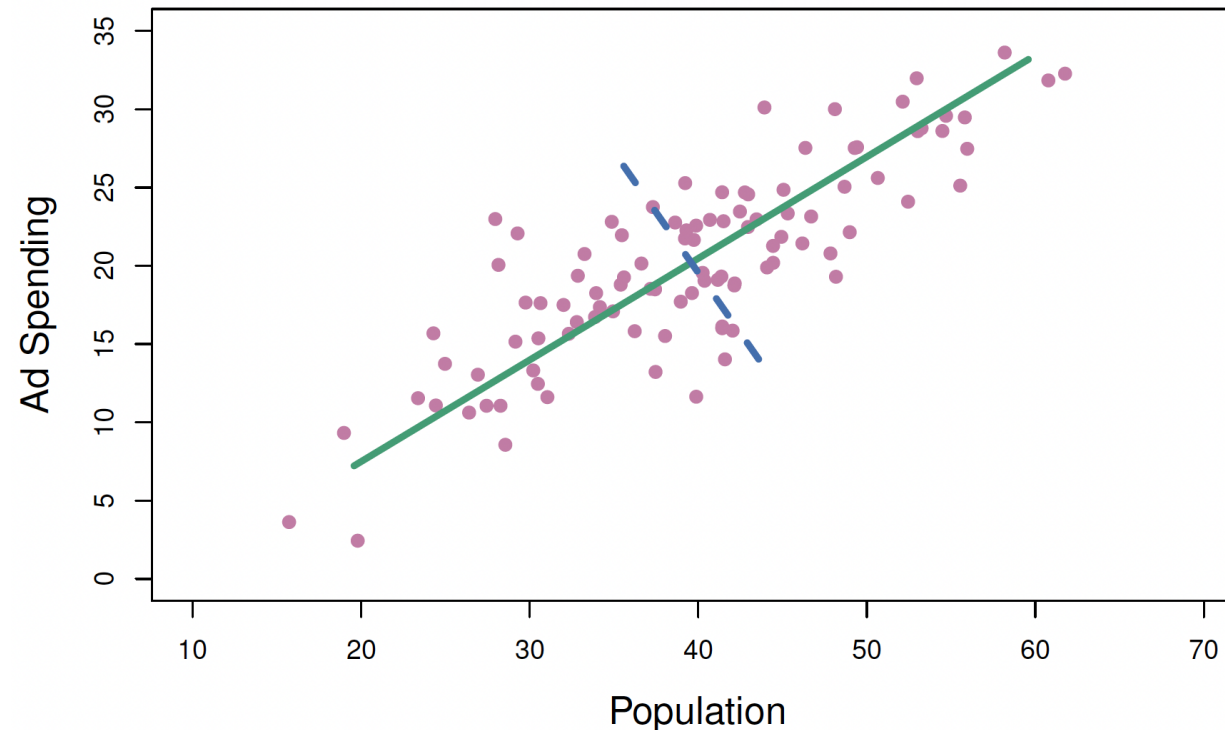
5.4 Dimension Reduction Methods

Principal Component Regression

- Here, we apply principal components analysis (PCA) (discussed in more detail in topic 9) to define the linear combinations of the predictors for use in our regression.
- The original predictors, p , are transformed into components M , where $M < p$.
- The first principal component is the linear combination of the variables with the largest variance.
- The second principal component has the second largest variance, subject to being uncorrelated with the first.
- And so on.
- Hence, with many correlated original variables, we replace them with a small set of principal components that capture their joint variation.

5.4 Dimension Reduction Methods

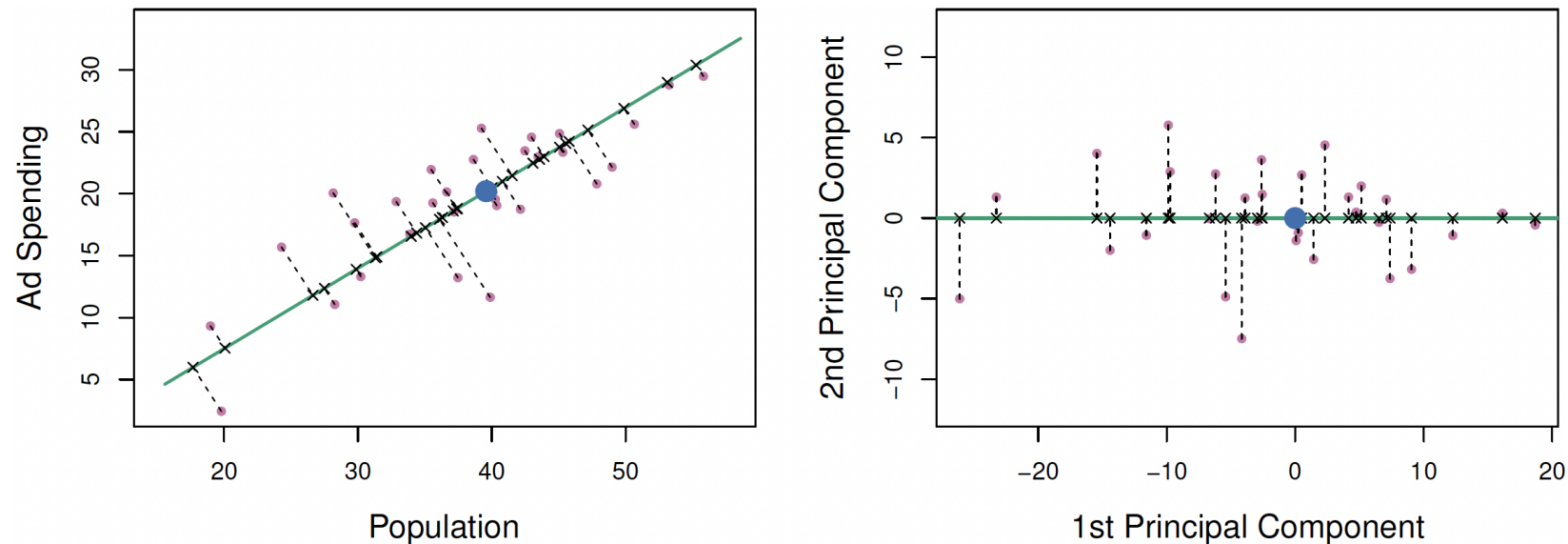
Principal Component Regression – continued



The population size (**pop**) and ad spending (**ad**) for 100 cities are in purple circles. The green solid line indicates the first principal component and the blue dashed line indicates the second principal component.

5.4 Dimension Reduction Methods

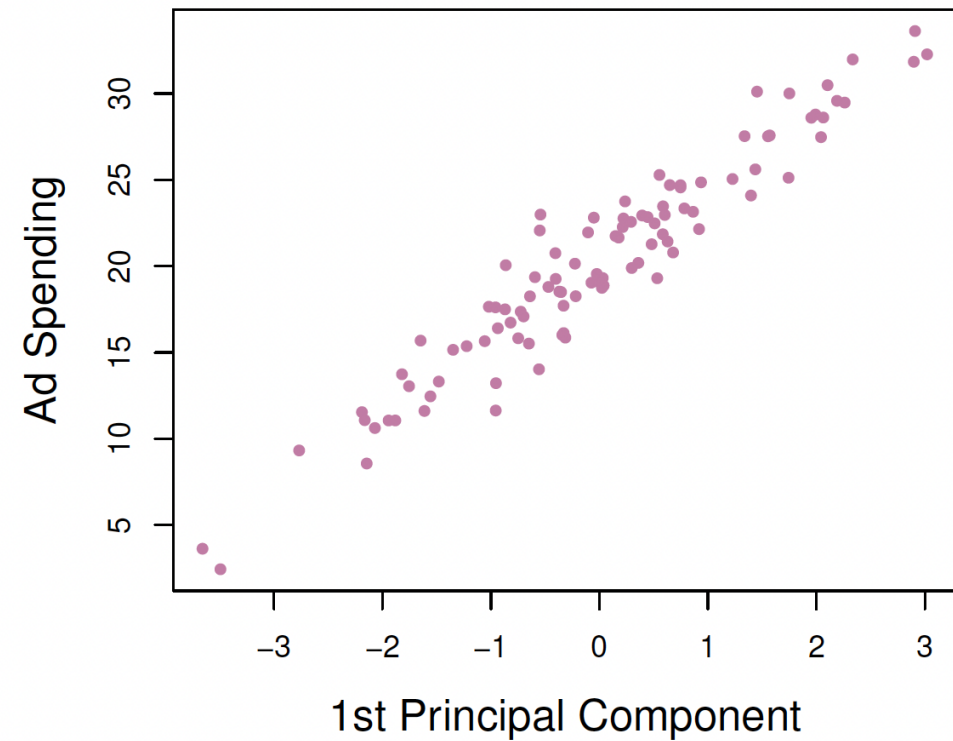
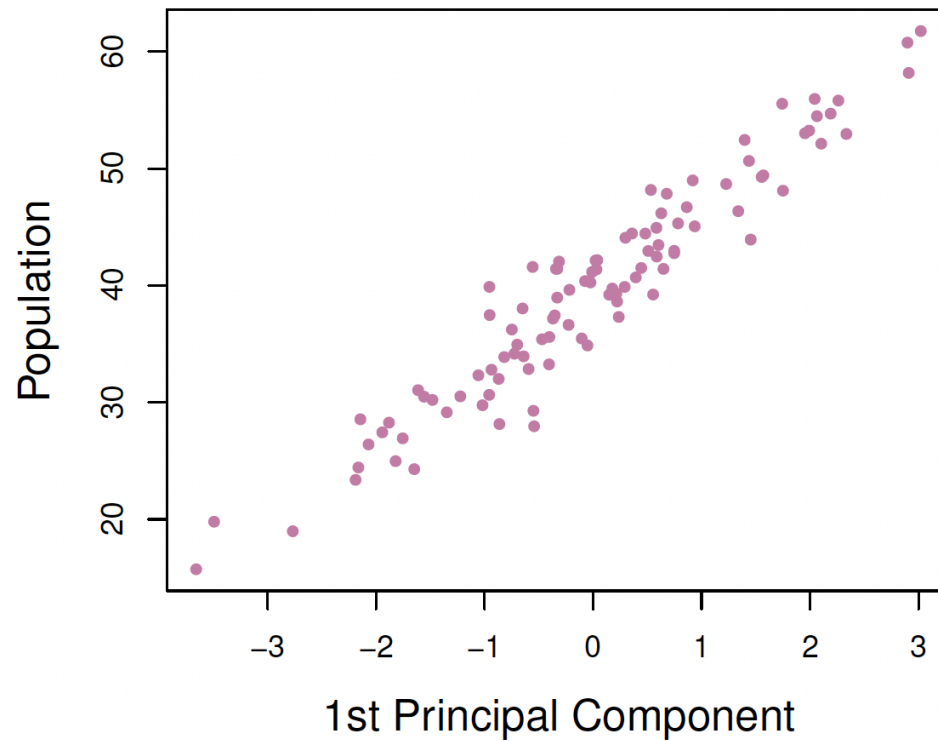
Principal Component Regression – continued



A subset of the advertising data. *Left:* The first principal component, chosen to minimize the sum of the squared perpendicular distances to each point, is shown in green. These distances are represented using the black dashed line segments. *Right:* The left-hand panel has been rotated so the first principal component lies on the x-axis.

5.4 Dimension Reduction Methods

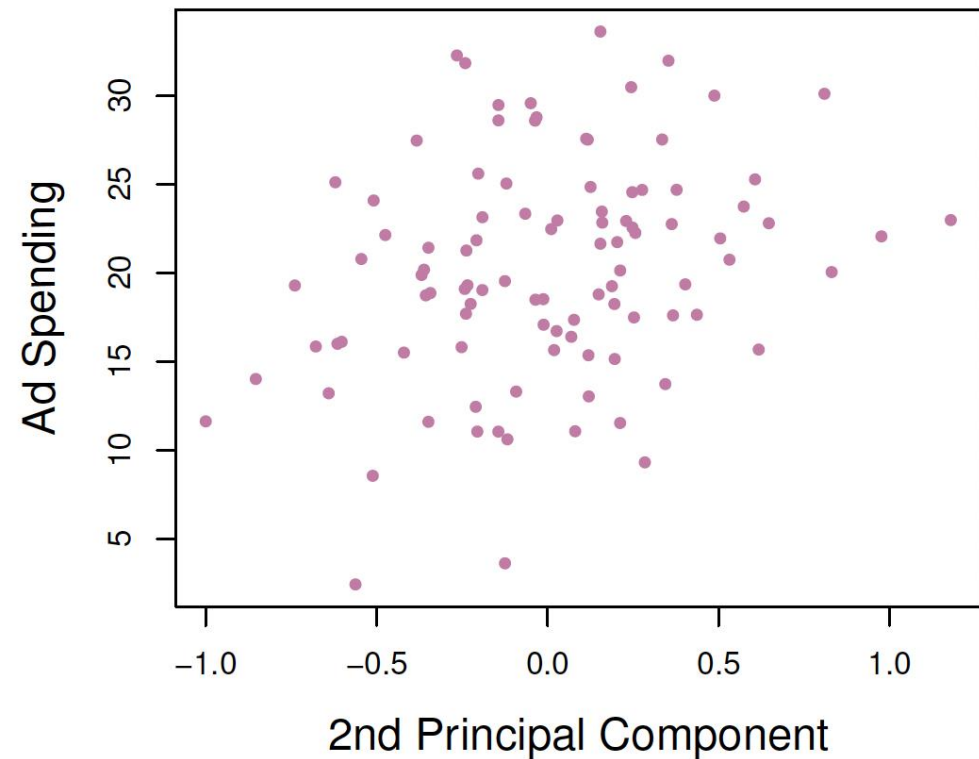
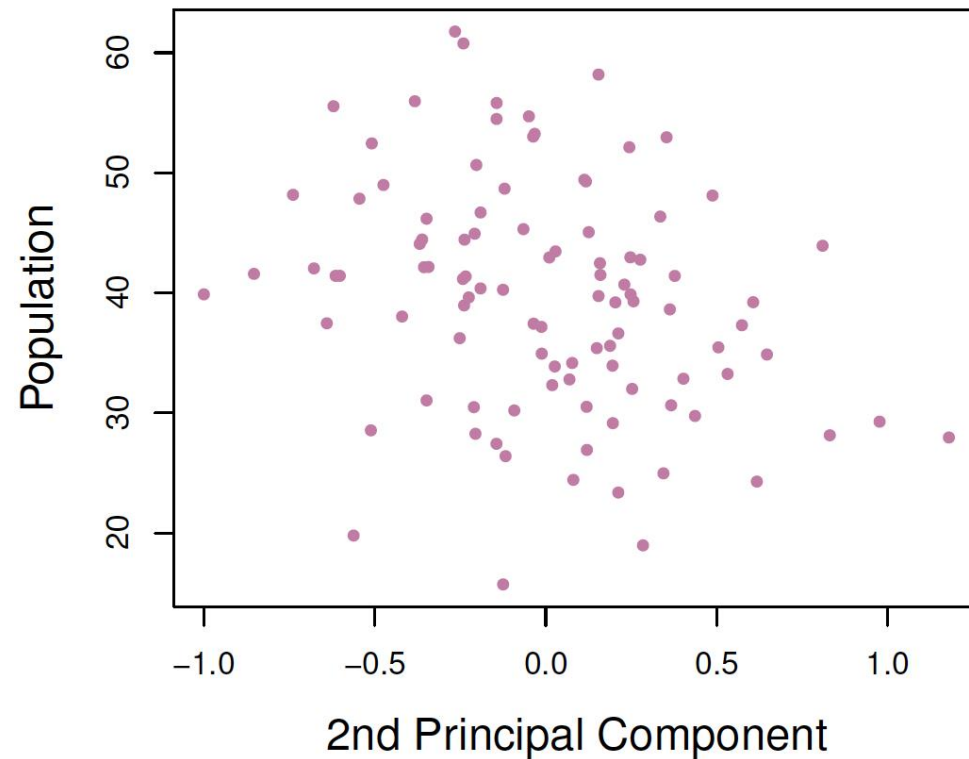
Principal Component Regression – continued



Plots of the first principal component scores versus **pop** and **ad**. The relationships are strong.

5.4 Dimension Reduction Methods

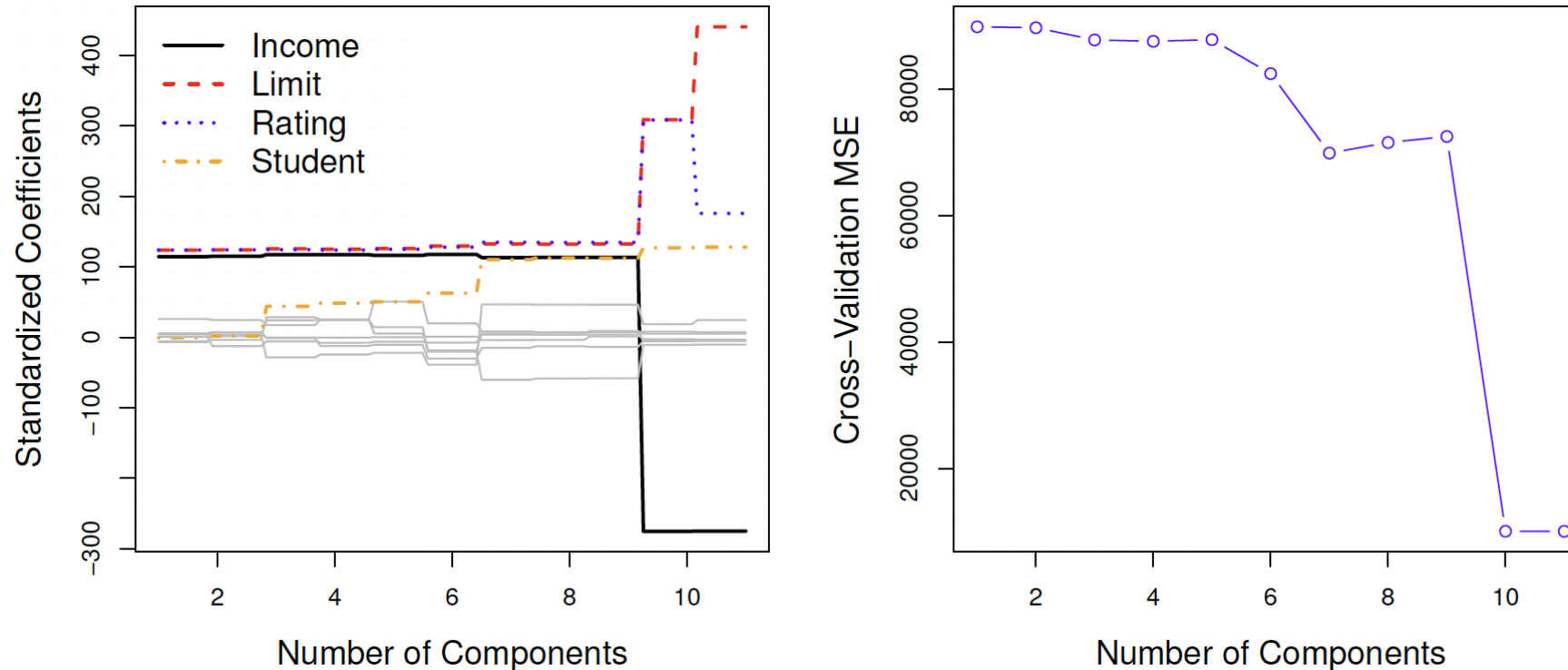
Principal Component Regression – continued



Plots of the second principal component scores versus **pop** and **ad**. The relationships are weak.

5.4 Dimension Reduction Methods

Principal Component Regression – continued



Left: PCR standardized coefficient estimates on the Credit data set for different values of M . **Right:** The 10-fold cross-validation MSE obtained using PCR, as a function of M .

5.4 Dimension Reduction Methods

Principal Component Regression – continued

- In summary, PCR identifies linear combinations or directions, that best represent the predictors $X_1, X_2, \dots, X_p$.
- These directions are identified in an unsupervised way since the response Y is not used to help determine the principal component directions.
- That is, the response does not supervise identifying the principal components.
- Consequently, PCR suffers from a potentially serious drawback.
- There is no guarantee that the directions that best explain the predictors will also be the best directions to use for predicting the response.

5.4 Dimension Reduction Methods

Partial Least Squares

- Like PCR, PLS is a dimension reduction method.
- But unlike PCR, PLS identifies these new features in a supervised way.
- It uses the response Y to identify new features that approximate the old features well and are also related to the response.
- Roughly speaking, the PLS approach attempts to find directions that help explain both the response and the predictors.
- In practice, PLS often isn't better than ridge or PCR.
- Supervision can reduce bias (vs PCR), but can also increase variance.

5.4 Dimension Reduction Methods

Partial Least Squares – continued

- After standardizing the p predictors, PLS computes the first direction by setting each direction value equal to the coefficient from the simple linear regression of Y onto X_j .
- One can show that this coefficient is proportional to the correlation between Y and X_j .
- Subsequent directions are found by taking residuals and then repeating the above prescription.

Summary

- Subset selection involves identifying a subset of the p predictors that are believed to be related to the response and fitting a least squares model to the reduced set of predictors.
- Shrinkage involves fitting a model involving all p predictors.
- However, the estimated coefficients are shrunk towards zero relative to the least squares estimates. Shrinkage reduces variance and may perform variable selection.
- Dimension reduction involves projecting all p predictors into an M -dimensional subspace, where $M < p$.
- Later, we will return to PCA in more detail.

Further Reading

- Chapter 6 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer
<https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 6 - Introduction to Statistical Learning Using R Book Club:
<https://r4ds.github.io/bookclub-islr/linear-model-selection-and-regularization.html>
- Summary of Chapter 6: <https://www.bijenpatel.com/guide/islr/linear-model-selection-regularization/>
- Best subset selection - <https://quantifyinghealth.com/best-subset-selection/>
- Stepwise selection - <https://quantifyinghealth.com/stepwise-selection/>

End of Topic

Thank you
Any questions?